NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 1

CANDIDATE NAME

CLASS

BIOLOGY

8876/02

Paper 2 Structured and Free-response Questions

13 September 2018

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and class in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do no use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
Total	

This document consists of **15** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

1 Cholesterol is synthesised in the smooth endoplasmic reticulum (SER) in liver cells by a series of enzyme-catalysed reactions.

Within the SER, molecules of cholesterol and triglycerides are surrounded by proteins and phospholipids to form lipoproteins. These lipoprotein particles enter the Golgi apparatus where they are packaged into vesicles and pass to the blood.

Fig. 1.1 is an electron micrograph of part of a liver cell showing lipoprotein particles within the Golgi apparatus.

Fig. 1.1

(a) Name structure T in Fig. 1.1 and state its role in liver cells.

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.... [2]

(b) Suggest why cholesterol is packaged into lipoproteins before release from liver cells into the blood.

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.... [1]

(c) Cholesterol is also packaged into vesicles by the SER and then secreted from the cell into small fluid-filled spaces between the liver cells. These spaces form ducts that drain into the gall bladder to form bile.

Explain how cholesterol is secreted into ducts, such as the duct in Fig. 1.1.

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.... [3]

(d) Both the Golgi body and the rough endoplasmic reticulum are part of the internal network of membranes in cells.

Outline structural features shown in Fig. 1.1 that identify **G** as the Golgi body and not the rough endoplasmic reticulum.

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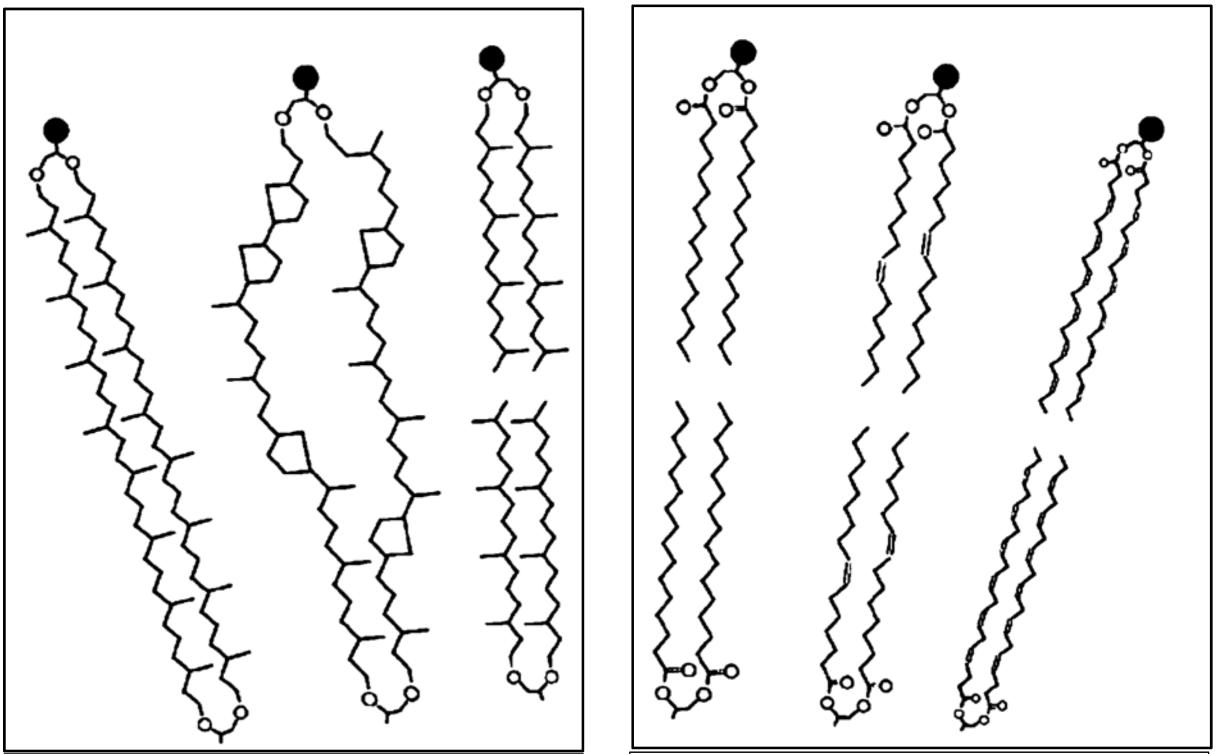
.... [2]

Cholesterol is a major component of all membranes. The concentration of cholesterol largely varies between membranes of different cells and tissues. There are other differences in the chemical composition of cell membranes in different organisms, such as the type of fatty acid chains in phospholipids.

Fig. 1.2 shows the structure of the phospholipids in the membranes of Organism A, which is an extreme thermophile (live in extremely high temperature places like hot springs), and Organism B, which live in normal environment (non-thermophile).



Organism A	**Organism B**

Fig. 1.2

(e) **(i)** With reference to Fig. 1.2, other than the presence of side branches and rings, state two structural differences between the phospholipids of Organism **A** and **B**.

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(ii) Suggest how the differences stated in **(e)(i)** enable Organism **A** to thrive in environments with extreme high temperature condition.

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[2]

[Total: 12]

2 Chitinases are enzymes synthesized by bacteria, fungi, yeasts, plants, that can degrade chitin into low molecular weight, soluble and insoluble oligosaccharides. Chitin is a modified polysaccharide found in a number of different organisms, for example in fungal cell walls and the hard outer skeletons of insects.

Chitinase is made up of 825 amino acids. Fig 2.1 shows the arrangement of some of the conserved amino acids found close together in the active site of chitinase. Fig. 2.2 shows the structure of a single chitinase molecule.

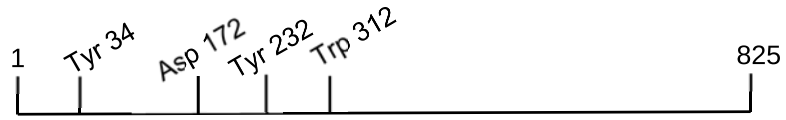

Fig. 2.1

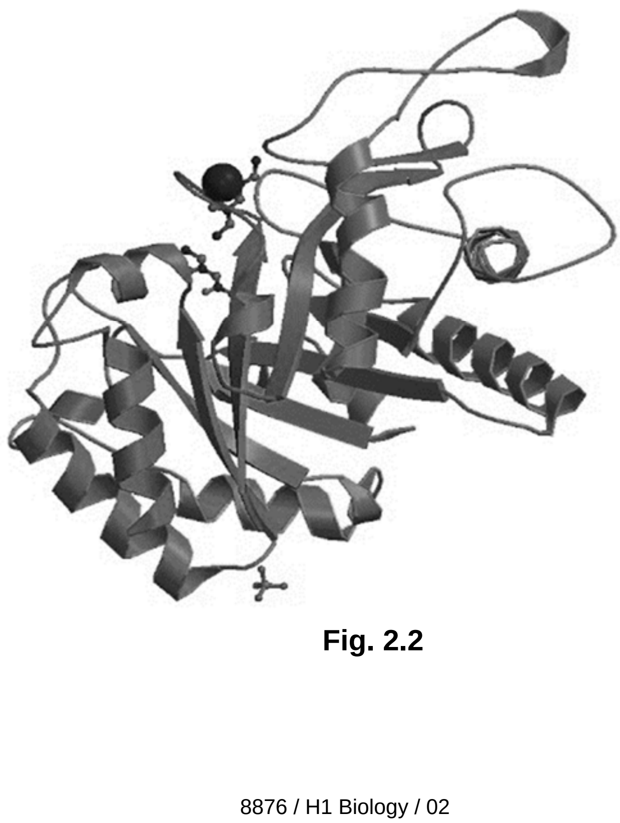

Fig. 2.2

(a) With reference to Fig 2.2, describe how the amino acid residues at different positions may be brought together when chitinase is synthesized.

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.... [3]

Chitin and the products of chitin hydrolysis have many useful medical and environmental applications. Chitinase enzymes can be used commercially to hydrolyse chitin. Enzyme stability and activity are important considerations in technological applications of chitinase.

Fig. 2.3 is a graph showing the effects of temperature on chitinase extracted from a soil bacterium. The relative activity of the enzyme was measured at different temperatures, with 100% representing maximum enzyme activity.

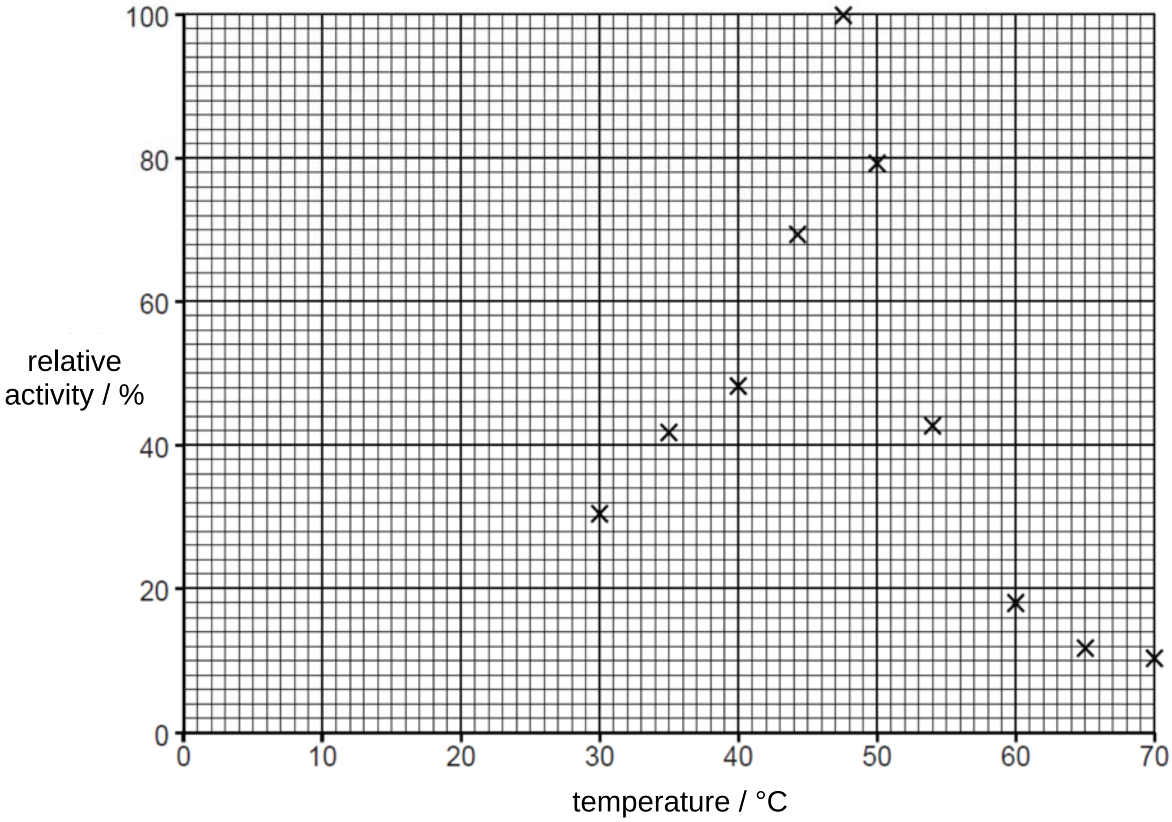

Fig. 2.3

(b) **(i)** With reference to Fig. 2.3, state the optimum temperature for the chitinase enzyme.

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Fig. 2.4 is a graph showing how temperature affects the stability of chitinase. The activity of the enzyme was measured over a time period of 72 hours at each of five different temperatures.

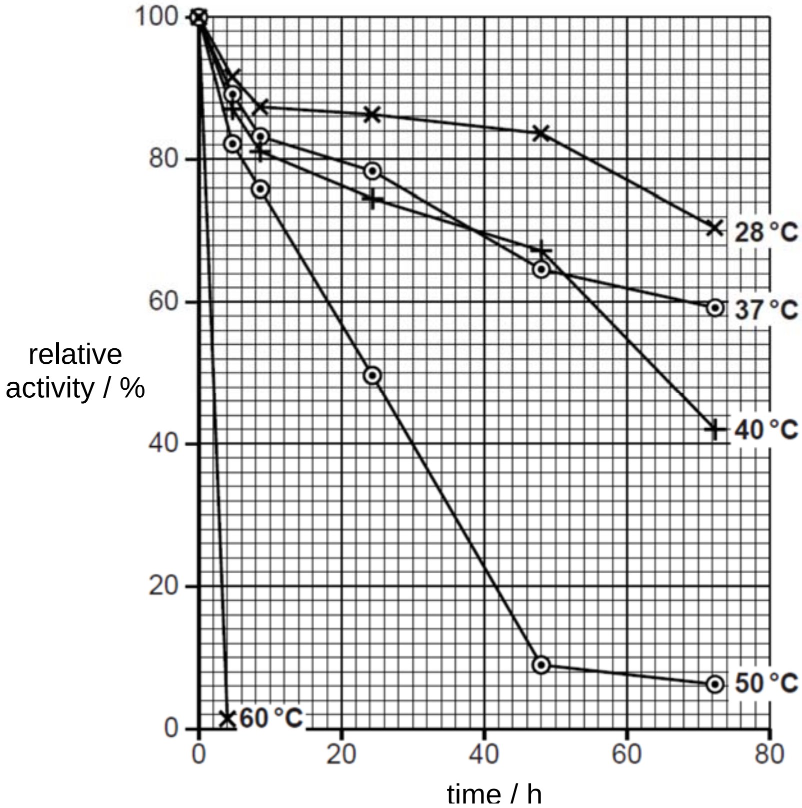

Fig. 2.4

(ii) With reference to Fig. 2.3 and Fig. 2.4, describe **and** discuss the effect of temperature on chitinase activity **and** stability.

[5]
[Total: 9]

3 In mice, fur colour is controlled by a gene with multiple alleles. These alleles are listed below in no particular order.

black and tan = C^{bt} yellow = C^{y}
agouti = C^{a} black = C^{b}

(a) Explain how multiple alleles arise.

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.... [2]

(b) Suggest explanations for the results of the following crosses between mice.

(i) Mice with agouti fur crossed with mice with black fur may produce all agouti offspring **or** some agouti and some black offspring.

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.... [2]

(ii) Crosses between heterozygous parents with the genotype $C^{y}C^{b}$ always produce a ratio of two yellow mice to one black mouse.

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(iii) Mice with yellow fur crossed with mice with black fur will produce one of the following outcomes:

- some yellow offspring and some agouti offspring
- some yellow offspring and some black and tan offspring
- some yellow offspring and some black offspring.

[2]

(c) A test cross is used to determine the genotype of an organism.

Describe how you would carry out a test cross to determine the genotype of a black and tan mouse.

[2]

[Total: 10]

4 Global warming has changed both the thickness and surface area of sea ice of the Arctic Ocean as well as the Southern Ocean that surrounds Antarctica. Sea ice is highly sensitive to changes in temperature.

Scientists have calculated a long-term mean for the surface area of sea ice in the Arctic and in the Southern Ocean around Antarctica. This mean value is used as a reference to examine changes in ice extent. The graph Fig. 4.1 shows the variations from this mean (zero line) over a period of time.

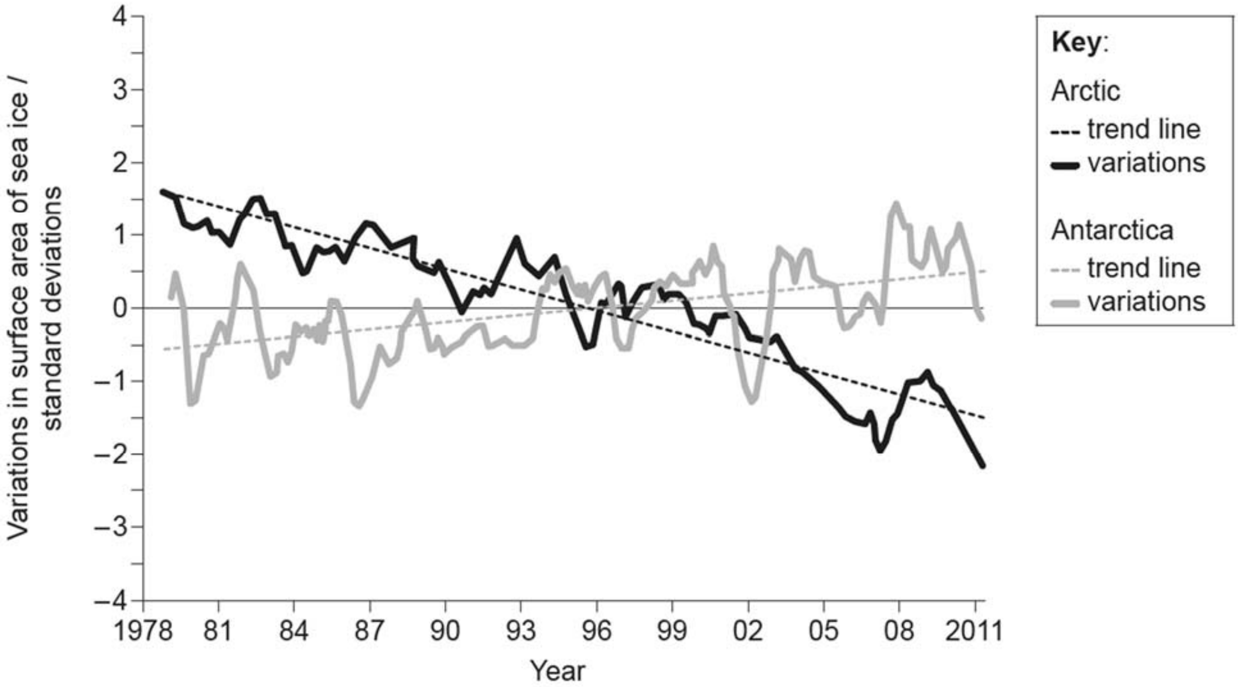

Fig. 4.1

(a) State the trend in the surface area of sea ice in the Southern Ocean around Antarctica.

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(b) Distinguish between changes in the surface area of sea ice in the Arctic and Antarctica.

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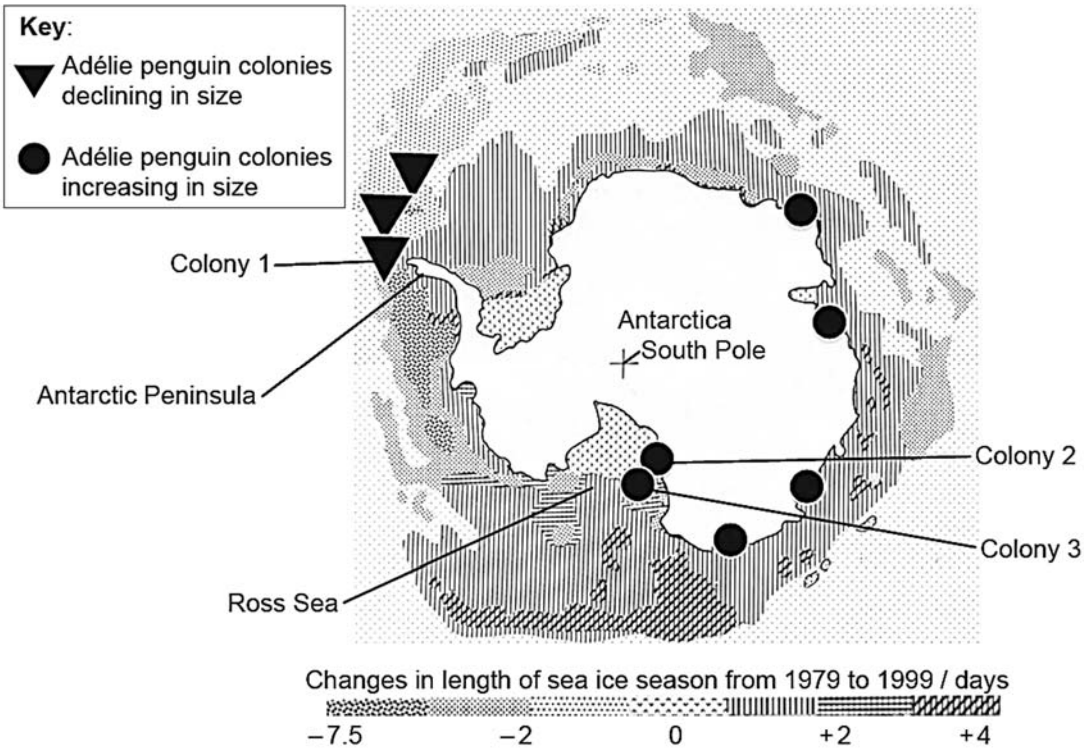
(c) Discuss the data as evidence of global warming.

[3]

Adélie penguins (*Pygoscelis adeliae*) are only found in Antarctica and need sea ice for feeding and nesting. Biologists are able to deduce how these penguins have responded to changes in their environment for the last 35 000 years, as the Antarctic conditions have preserved their bones and their nests. The image is a map of Antarctica and the surrounding Southern Ocean. It shows the trends in the length of the sea ice season (days of the year when sea ice is increasing) and the sites of nine Adélie penguin colonies.



[Source: Data sourced from the penguinscience.com website]

Fig. 4.2

(d) Describe the trends in the length of the sea ice season around the Antarctic Peninsula and in the Ross Sea.

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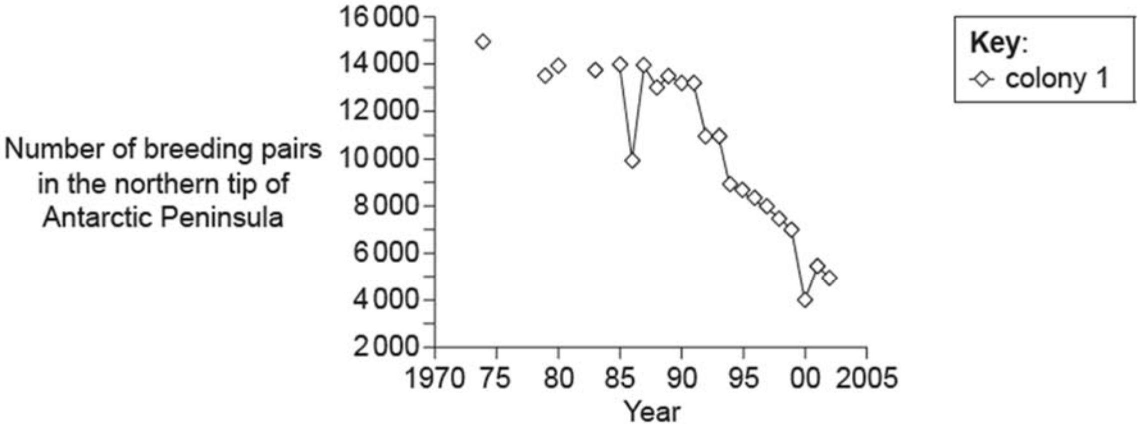
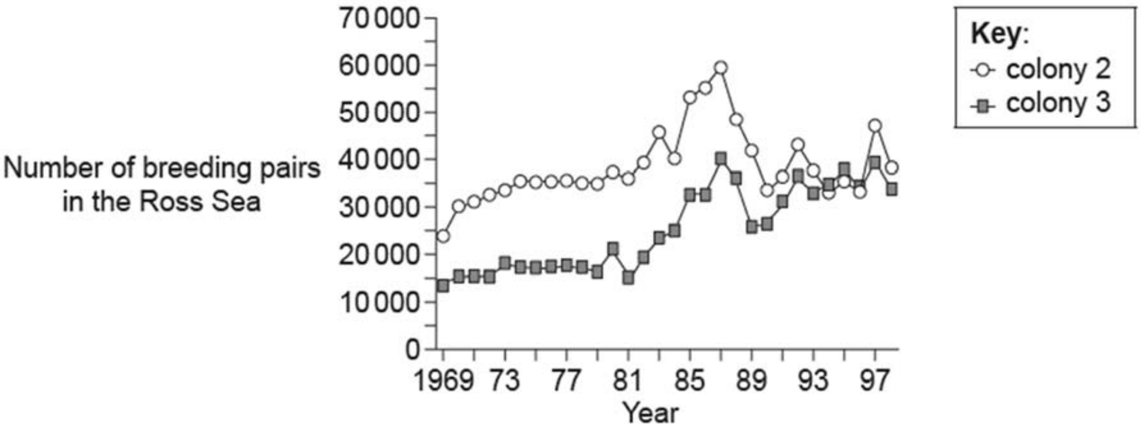
The graphs show the changes in penguin population in three of the colonies shown on the map.





[Source: Data sourced from: www.penguinscience.com/clim_change.php]

Fig. 4.3

(e) Analyse the trends in colony size of the Adélie penguins in relation to the changes in the sea ice.

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(f) Discuss the use of Adélie penguins in studying the effects of global warming.

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[Total: 14]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Outline the structural differences between typical prokaryotic and eukaryotic cells and explain how it relates to differences in gene expression. [6]

(b) Explain, with examples, how environmental factors act as forces of natural selection. [9]

[Total: 15]

6 **(a)** Explain how organisms grown from genetically identical zygotes can have different phenotypes. [6]

(b) Charles Darwin proposed that evolution occurs primarily by natural selection. However deleterious recessive alleles are not eliminated from population.
Describe and explain how these alleles remain in the population. [9]

[Total: 15]